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Roll no.:

IIIrd Semester- B.Tech Course work
END SEMESTER EXAMINATION, NOVEMBER-DECEMBER, 2023

Course code: CMMTC04

Course title: Engineering Mathematics

Time: 3 hours

Maximum Marks. 50

Note: Missing data/information(if any), may be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO																								
Q1	Attempt any 2 parts of the following																										
1a	Solve $y'' + 3y' + 2y = \delta(t - 1), y(0) = y'(0) = 0$.	5	CO																								
1b	Find Laplace Transform of the following function : a) $f(t) = \int_0^\infty e^{-2t} t \sin^3 t dt$ b) $f(t) = \frac{1 - \cos t}{t^2}$	5	CO																								
1c	Find the Laplace Transform of the Periodic function of period T $f(t) = \frac{kt}{T}, 0 < t < T$	5	CO																								
Q2	Attempt any 2 parts of the following																										
2a	Find the sine half range expansions of the function : $f(x) = \begin{cases} \frac{2k}{L}x & 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L - x) & \frac{L}{2} < x < L \end{cases}$	5	CO																								
2b	Find a Fourier series to represent the function $f(x) = \begin{cases} 0 & -\pi < x < 0 \\ \frac{1}{4}(\pi x) & 0 < x < \pi \end{cases}$	5	CO																								
2c	A machine completes its cycle of operation every time as certain pulley completes a revolution. The displacement $f(x)$ of a point on a certain portion of the machine is given in the table given below for twelve positions of the pulley, x being the angle in degree turned through by the pulley. Find a Fourier series to represent $f(x)$ for all values of x . <table border="1" style="margin: 10px auto;"><tr><td>x</td><td>30°</td><td>60°</td><td>90°</td><td>120°</td><td>150°</td><td>180°</td><td>210°</td><td>240°</td><td>270°</td><td>300°</td><td>330°</td></tr><tr><td>$f(x)$</td><td>7.976</td><td>8.026</td><td>7.204</td><td>5.676</td><td>3.674</td><td>1.764</td><td>0.552</td><td>0.262</td><td>0.904</td><td>2.492</td><td>4.736</td></tr></table>	x	30°	60°	90°	120°	150°	180°	210°	240°	270°	300°	330°	$f(x)$	7.976	8.026	7.204	5.676	3.674	1.764	0.552	0.262	0.904	2.492	4.736	5	CO
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$f(x)$	7.976	8.026	7.204	5.676	3.674	1.764	0.552	0.262	0.904	2.492	4.736																
Q3	Attempt any 2 parts of the following																										
3a	Solve $px(z - 2y^2) = (z - qy)(z - y^2 - 2x^3).$	5	CO																								
3b	Let $u(x, t)$ be the solution of the initial boundary value problem $\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, -\infty < x < \infty, t > 0, u(x, 0) = x^4, u_t(x, 0) = \sin x$. Then find the value of $u(2, 2)$.	5	CO																								
3c	Solve the equation $u_{xx} + u_{yy} = 0, 0 < x < l, 0 < y < a, u(x, 0) = \sin \frac{n\pi x}{l}, u(x, a) = 0, u(0, y) = 0, u(l, y) = 0.$	5	CO																								

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Q4	Attempt any 2 parts of the following		
4a	Show that the function defined by $f(z) = \sqrt{ xy }$ satisfies Cauchy-Riemann equation at the origin but is not analytic at that point.	5	CO4
4b	Evaluate the line integral $\int_C z^2 dz$ where C is the boundary of a triangle with vertices $0, 1 + i, -1 + i$ clockwise.	5	CO4
4c	Evaluate $\oint_C \frac{12z - 7}{(z - 1)^2(2z + 3)} dz,$ where C is the circle $ z + i = \sqrt{3}$	5	CO5
Q5	Attempt any 2 parts of the following		
5a	Find the constant a, b, c so that $v = (x + 2y + az)\hat{i} + (bx - 3y - z)\hat{j} + (4x + cy + 2z)\hat{k}$ is irrotational and express the v as the gradient of the scalar function ϕ .	5	CO5
5b	Evaluate $I = \int_C \frac{e^y}{x} dx + (e^y \log x + x) dy$ where C is the positively oriented boundary of the region enclosed by $y = 1 + x^2; y = 2, x = \frac{1}{2}$.	5	CO5
5c	Let $a > 0$ and $S = \{(x, y, z) \in R^3 : x^2 + y^2 + z^2 = a^2\}$. Evaluate the surface integral $\int \int_S (x^3 \hat{i} + y^3 \hat{j} + z^3 \hat{k}) \frac{(x \hat{i} + y \hat{j} + z \hat{k})}{a} dS.$	5	CO5